LIVEVSR: REVIVING LIVE PHOTOS VIA COVER-GUIDED VIDEO SUPER-RESOLUTION

Anonymous authors
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ABSTRACT

Live Photo pairs a high-quality cover image with a lower-quality video of the same moment. Existing cover-guided methods mainly enhance individual frames, while complete video restoration has received less attention. Two challenges hinder this task: uncertain cover-video correspondence and residual degradation in the cover. To address these challenges, we propose LiveVSR, a cover-guided one-step diffusion framework for complete Live Photo super-resolution. We combine continuous reference features with a discrete visual prior to better use the cover. First, we build an in-context framework to jointly model cover and video tokens through bidirectional attention. Flow-guided local retrieval combines motion cues with content-based reference matching. Meanwhile, we introduce correspondence-guided codebook injection. A codebook pretrained on high-quality images provides discrete cover features. These features are transferred through contextualized correspondences to complement continuous reference guidance. Training with clean and mildly degraded covers further accounts for reference imperfections. We compare the proposed LiveVSR with conventional VSR, reference-guided VSR, and Live Photo image enhancement methods. Experiments on synthetic benchmarks and real Live Photos demonstrate favorable visual quality, reconstruction fidelity, and temporal consistency. Code will be released.

1 INTRODUCTION

Live Photos record expressions, gestures, and scene changes around a photograph. Each capture contains a high-quality cover image and a short video of the surrounding moments. However, the two observations follow different imaging and compression pipelines (Xue et al., 2026; Lou et al., 2026). The video thus often loses details that remain clear in the cover. This quality gap motivates us to reconstruct the complete video using the native cover. We term this task **cover-guided Live Photo super-resolution**. As illustrated in Fig. 1, we aim to improve visual quality using appearance details from the same capture. The restored sequence should also maintain temporal consistency.

Recent work has explored using the cover to enhance a selected video frame. Some methods (Xue et al., 2026) use reference-guided diffusion to transfer details. Others (Lou et al., 2026) recover details by fusing neighboring frames with cover guidance. Although these methods improve visual quality, their output remains a single image. Restoring the complete video imposes additional demands: reference details must remain consistent across frames to avoid artifacts and flicker.

Video super-resolution (VSR) offers a natural starting point. Convolutional methods reconstruct details through multi-frame aggregation and propagation (Jo et al., 2018; Li et al., 2020; Chan et al., 2021; 2022a;b). Transformer-based methods further model long-range dependencies (Liang et al., 2022; 2024; Zhang & Yao, 2024). Diffusion-based methods exploit pretrained image and video priors for detail restoration (Zhou et al., 2024; Yang et al., 2024; He et al., 2024; Xie et al., 2025; Wang et al., 2025b; Chen et al., 2025; Wang et al., 2026). However, these methods do not exploit the native cover to recover details missing from the video.

Reference-based VSR explores auxiliary visual guidance. Some methods use synchronized videos or images captured at different focal lengths (Lee et al., 2022; Zou et al., 2025). Others transfer details from a single reference image or enhanced keyframes (Kim et al., 2023; Yu et al., 2026). In Live Photos, a static reference is already available. It provides a view similar to the video but depicts only one moment. Applying reference guidance to this setting requires reliable correspondence and consistent detail transfer throughout the sequence.

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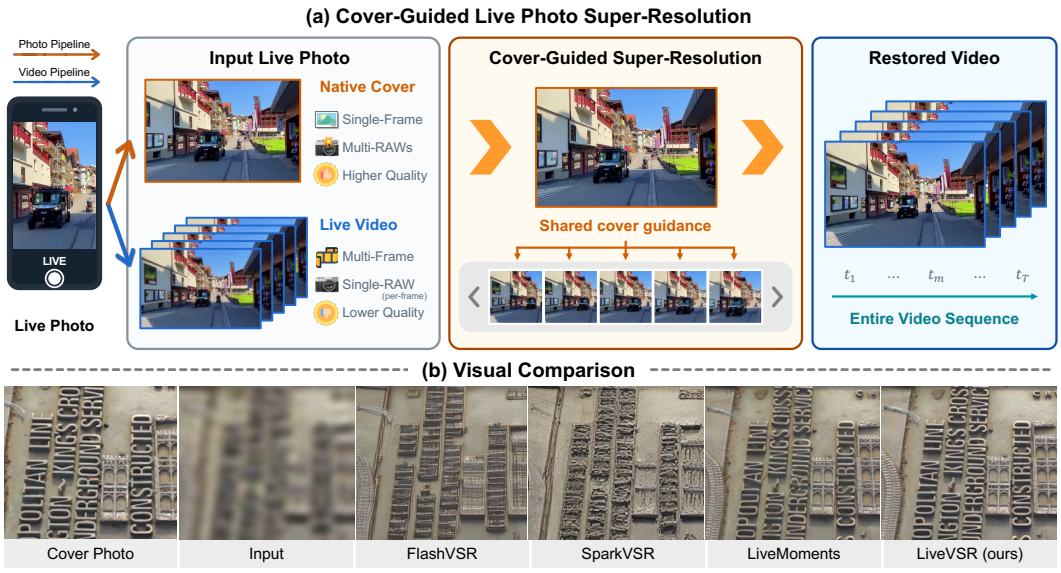


Figure 1: Task illustration and visual comparison for cover-guided Live Photo super-resolution. (a) The native cover guides restoration of the complete video sequence. (b) LiveVSR recovers clearer structures and finer details than competing methods.

Two challenges arise in this task. **(1) Matching the cover to a dynamic sequence.** Motion, occlusion, and imaging differences make cover-video correspondence uncertain. The cover captures only one instant, so some video regions may have no visible counterpart. Reference matching must therefore use motion cues while adapting to changes in visibility and appearance. **(2) The cover frame may be degraded.** Although the cover is typically clearer than the accompanying video, it is still subject to imperfect capture conditions and image processing. Low-light noise, motion blur, and processing artifacts may therefore remain in the cover. Direct feature transfer can propagate these degradations into the video. Reference guidance should retain useful appearance details while limiting the influence of residual cover degradation.

To address these challenges, we propose **LiveVSR**, a cover-guided framework for complete Live Photo restoration. We adapt a pretrained video diffusion backbone (Wang et al., 2025a) for one-step restoration. Our framework combines continuous reference features with a discrete visual prior to better use and integrate cover information. First, we build an **in-context framework** with bidirectional attention between cover and video tokens. Flow-guided local retrieval refines cover matching using motion cues. A confidence-weighted mixture combines local motion correspondences with global content-based matching of reference features.

Meanwhile, we introduce **correspondence-guided codebook injection**. A codebook pretrained on high-quality images supplies a discrete cover representation. We reuse contextualized video queries and cover keys to select reference locations. Their code features are injected through a residual branch to supplement continuous guidance. Training with clean and mildly degraded covers further accommodates variation in native reference quality.

We evaluate LiveVSR on real Live Photos. We compare with conventional VSR, reference-based VSR, and Live Photo image enhancement methods. As shown in Fig. 1, LiveVSR achieves better visual quality and recovers more details. Our contributions are summarized as follows:

- We formulate cover-guided Live Photo super-resolution and develop LiveVSR. It uses the native cover to restore the complete video through one-step diffusion.
- We introduce an in-context framework and correspondence-guided codebook injection. They combine continuous and discrete reference features to use cover information.
- We conduct experiments on synthetic benchmarks and real Live Photos. LiveVSR outperforms existing methods in visual quality, reconstruction fidelity, and temporal consistency.

2 Related Work

2.1 Video Super Resolution

Video super-resolution (VSR) aims to reconstruct high-resolution videos from low-resolution sequences while maintaining temporal consistency. Early convolutional methods exploit neighboring frames through spatiotemporal modeling and feature aggregation (Jo et al., 2018; Li et al., 2020). Beyond local aggregation, BasicVSR propagates features bidirectionally (Chan et al., 2021), while Transformer architectures model cross-frame dependencies through attention (Liang et al., 2022; 2024). The shift toward real-world inputs also motivates paired video datasets that better capture practical degradations (Yang et al., 2021; Wang et al., 2023). At the model level, input cleaning in RealBasicVSR (Chan et al., 2022b) and robust attention designs (Zhang & Yao, 2024) help limit artifact propagation. To recover missing details, diffusion-based methods draw on pretrained generative priors. To adapt image priors to video, Upscale-A-Video combines temporal layers with latent propagation (Zhou et al., 2024). Related approaches use motion guidance and temporal attention to improve consistency across frames (Yang et al., 2024; Li et al., 2025; Chen et al., 2026b). Video diffusion models, including STAR, provide joint spatial and temporal priors for restoration (He et al., 2024; Xie et al., 2025; Wang et al., 2025b; Chen et al., 2025; 2026a). DOVE and SeedVR2 further reduce sampling cost through one-step restoration (Chen et al., 2025; Wang et al., 2026). These developments provide the generative and temporal foundation for our work. In Live Photos, the native cover offers an additional appearance observation from the same capture. We combine this photographic evidence with learned video priors to restore the full recorded sequence.

2.2 Reference-Guided Image and Video Restoration

Reference-guided restoration uses high-quality references to supply details missing from degraded inputs. Image methods address the resulting correspondence problem through feature matching, texture transfer, and robust alignment (Zhang et al., 2019; Yang et al., 2020; Jiang et al., 2021; Cao et al., 2022). Retrieved images can also ground diffusion-based restoration by supplying external texture cues (Guo et al., 2024). In video restoration, the source and temporal coverage of the reference become important. Multi-camera approaches exploit synchronized reference videos (Lee et al., 2022; Zou et al., 2025), while ERVSR uses a single reference image to reduce computation (Kim et al., 2023). SparkVSR conditions video diffusion on sparse keyframes, which can be enhanced before their details are propagated (Yu et al., 2026). Our work builds on these principles of reference matching and temporal modeling for cover-guided video restoration.

We focus on Live Photos, where a native cover photograph accompanies a lower-quality video of the same scene. The cover provides captured appearance details, but differs from the video in imaging pipeline and represents only one instant. Closely related work uses photo guidance to restore a reselected key photo, as in LiveMoments (Xue et al., 2026) and the Live2K benchmark (Lou et al., 2026). Live2K also incorporates neighboring frames, yet the restoration target remains a single image. Our task extends this target to the full recorded sequence. This requires consistent transfer of reference details across frames as motion and occlusion change their visibility. We therefore jointly model the cover and video to recover appearance details while retaining the recorded dynamics.

3 Method

3.1 Task Formulation

Given a Live Photo, let $\mathbf{X}_{1:T} = \{\mathbf{X}_t\}_{t=1}^{T}$ denote its low-quality video, where $\mathbf{X}_t \in \mathbb{R}^{H\times W\times 3}$. The native cover photograph from the same capture is denoted by $\mathbf{R} \in \mathbb{R}^{H_R\times W_R\times 3}$. Our goal is to restore the full video at a higher resolution using appearance information from the cover. The restored sequence should achieve visual quality comparable to the cover, preserve the scene content and recorded motion, and remain temporally consistent. We formulate this task as learning a restoration mapping F_θ from the input video and its paired native cover:

$$\mathbf{Y}_{1:T} = F_\theta(\mathbf{X}_{1:T}, \mathbf{R}), \qquad \mathbf{Y}_t \in \mathbb{R}^{sH\times sW\times 3}, \tag{1}$$

where s is the super-resolution scale factor and θ denotes the model parameters. The output retains the input's frame count and temporal order. We treat the native cover as a relatively high-quality reference observation that may contain residual degradations.

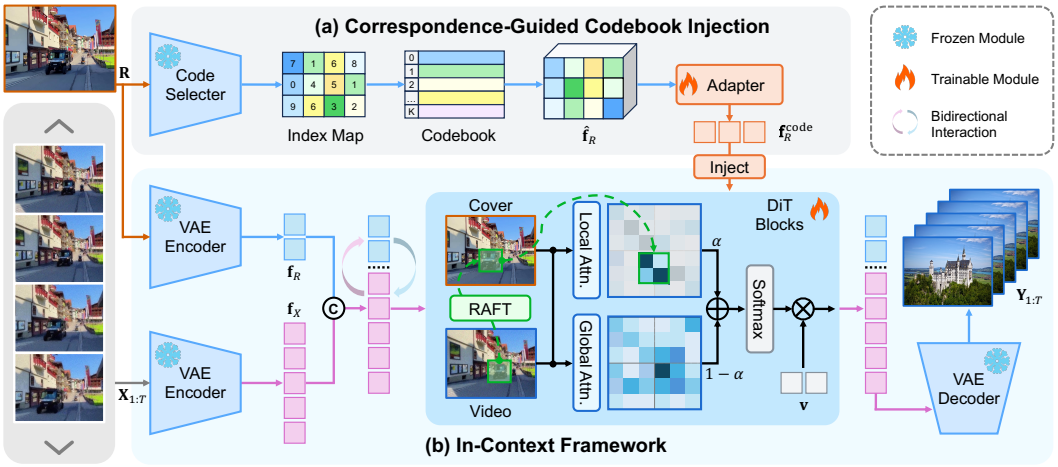


Figure 2: Overview of LiveVSR. (a) Codebook injection supplies discrete cover features. (b) The in-context framework combines cover and video tokens with flow-guided retrieval.

3.2 Overview

We build LiveVSR on Wan (Wang et al., 2025a), a video diffusion model pretrained with flow matching. Its DiT is pretrained to predict the velocity $\boldsymbol{\epsilon} - \mathbf{z}_0$, where $\mathbf{z}_0$ is the clean video latent and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Following previous one-step restoration methods (Chen et al., 2025), we directly use the encoded LQ video as $\mathbf{z}_\tau$ and estimate the clean latent through the reverse update:

$$\hat{\mathbf{z}}_0 = a_\tau \mathbf{z}_\tau - b_\tau \mathbf{v}_\theta(\mathbf{z}_\tau, \tau), \qquad \mathbf{z}_\tau = \mathcal{E}(\mathbf{x}_{\mathrm{LQ}}), \tag{2}$$

where $\mathcal{E}$ is the video encoder, $\mathbf{x}_{\mathrm{LQ}}$ denotes the LQ video, $\tau \in [0, 1]$ is the normalized restoration timestep, and $\mathbf{v}_\theta$ denotes the DiT's velocity prediction. The coefficients are determined by the noise schedule; for our flow-matching backbone, $a_\tau = 1$ and $b_\tau = \tau$.

The cover provides detailed appearance information from the same capture, but using this static observation to guide video restoration requires accounting for changing content and motion. We address this through two complementary components for cover conditioning. The *in-context framework* (Sec. 3.3) jointly models cover and video features and incorporates flow-guided local retrieval, combining motion cues with content-based reference matching. The *correspondence-guided codebook injection* (Sec. 3.4) reuses the contextualized queries and keys to transfer discrete cover features into the video. These components are optimized with the restoration objectives described in Sec. 3.5.

3.3 In-Context Framework

We treat the cover as visual context that participates directly in the video DiT's self-attention. We encode the resized cover and upsampled video with a shared VAE encoder and patch projection, and add source-specific type embeddings to obtain cover tokens $\mathbf{f}_R$ and video tokens $\mathbf{f}_X$. We concatenate them along the sequence dimension to form the joint input:

$$\mathbf{f}_{\mathrm{in}} = [\mathbf{f}_R, \mathbf{f}_X]. \tag{3}$$

Here, brackets denote sequence concatenation. Each DiT block forms joint queries $\mathbf{q}$, keys $\mathbf{k}$, and values $\mathbf{v}$, with queries and keys retaining the backbone's normalization and positional encoding.

In unidirectional interaction, video tokens attend to both cover and video tokens, while cover tokens attend only to cover tokens. Bidirectional interaction additionally allows cover tokens to attend to video tokens. We adopt the bidirectional design so that both cover and video representations evolve with the shared joint context throughout the DiT blocks.

This joint attention establishes implicit cover-video correspondences through feature similarity. As scene content moves relative to the cover, degraded observations and repetitive textures can make the appropriate reference location ambiguous. We therefore use motion cues to provide local spatial guidance for retrieving corresponding reference features.

Flow-guided local retrieval. To reduce the influence of the cover-video quality gap on motion estimation, a frozen optical-flow estimator computes correspondences from each LQ frame to the middle LQ frame. We use these locations as an approximate spatial prior for cover retrieval. To account for VAE temporal compression, we retain separate correspondences for the RGB frames $\mathcal{T}_i$ associated with video token i. Each frame t defines a local cover neighborhood $\mathcal{N}_{i,t}$ around its mapped position. The shared queries and keys yield the original and local cover distributions:

$$\begin{aligned} \boldsymbol{\alpha}_i^{\text{soft}} &= \text{softmax}\left(\frac{\mathbf{q}_{X,i}\mathbf{k}_R^\top}{\sqrt{d}}\right), \\ \boldsymbol{\alpha}_{i,t}^{\text{local}} &= \text{softmax}_{\mathcal{N}_{i,t}}\left(\frac{\mathbf{q}_{X,i}\mathbf{k}_R^\top}{\sqrt{d}}\right). \end{aligned} \tag{4}$$

The first softmax normalizes over all cover tokens, whereas the second normalizes within $\mathcal{N}_{i,t}$ and is zero elsewhere. Forward–backward consistency and valid-region checks yield confidence weights $c_{i,t} \in [0,1]$. We mix these two distributions using the confidence weights as follows:

$$\widetilde{\boldsymbol{\alpha}}_i = \boldsymbol{\alpha}_i^{\text{soft}} + \frac{\lambda_{\text{f}}}{|\mathcal{T}_i|}\sum_{t\in\mathcal{T}_i} c_{i,t}\left(\boldsymbol{\alpha}_{i,t}^{\text{local}} - \boldsymbol{\alpha}_i^{\text{soft}}\right), \tag{5}$$

where $\lambda_{\text{f}} \in [0,1]$ controls the mixing strength. Invalid correspondences have $c_{i,t} = 0$ and are skipped, so fully invalid queries retain the original attention. We rescale the mixture by the cover's original total attention weight, preserving this total and all video-key weights. Confidence weighting and the global component of the mixture retain flexible content-based matching under uncertain flow. All cover queries retain their original joint attention. Projecting the joint-attention outputs yields $\mathbf{f}_R^{\text{attn}}$ and $\mathbf{f}_X^{\text{attn}}$. After the final block, only video tokens are used to predict the velocity and restore the video through the corresponding restoration update in Eq. 2.

3.4 Correspondence-Guided Codebook Injection

The continuous reference tokens from the in-context framework provide appearance context and matching features that evolve with the video's dynamics. The native cover may still contain residual degradations, which can also be reflected in these continuous features. We therefore complement them with a discrete cover representation drawn from a high-quality image codebook, providing additional appearance guidance along the backbone's evolving cover-video correspondences.

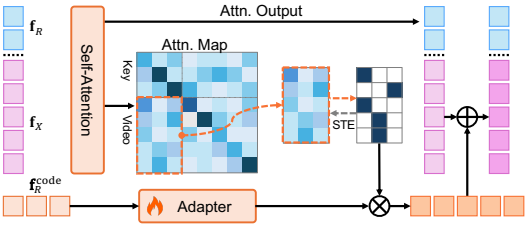


Figure 3: Codebook injection with hard reference retrieval. STE denotes the straight-through estimator used to train hard reference retrieval.

Discrete cover representation. We use a codebook $\mathcal{C}$ pretrained on high-quality images to represent the cover. The pretrained reference selector $\mathcal{S}$ selects a code index at each spatial position of the cover photograph R, and codebook lookup maps these indices to the corresponding pretrained code vectors. This yields a discrete feature map (Zhou et al., 2022; Chen et al., 2022):

$$\mathbf{F}_R^c = \mathcal{Q}_{\mathcal{C}}(\mathcal{S}(R)). \tag{6}$$

Here, $\mathcal{Q}_{\mathcal{C}}$ retrieves the code vectors in $\mathcal{C}$ using the selected indices, and the superscript c denotes the codebook branch. A learnable patch projection maps $\mathbf{F}_R^c$ to code tokens $\mathbf{f}_R^c$ on the same spatial grid as $\mathbf{f}_R$. Each cover position j therefore pairs a continuous feature $\mathbf{f}_{R,j}$ with a code feature $\mathbf{f}_{R,j}^c$.

Correspondence-guided value injection. The spatial alignment between the two cover representations lets us use continuous features to establish correspondence and code features to supply appearance values. Within each DiT block, we reuse the video queries $\mathbf{q}_{X,i}$ and cover keys $\mathbf{k}_{R,j}$ from the in-context framework. A learned value mapping converts the aligned code tokens $\mathbf{f}_R^c$ into code values $\mathbf{v}_R^c$. We denote the learned output projection by $\mathcal{P}$. To limit mixing across different cover locations in the code branch, we use *hard reference retrieval*: each video token retrieves the cover location with the highest matching score and uses its code value to form a residual:

$$\begin{aligned} j^*(i) &= \arg\max_j \langle \mathbf{q}_{X,i}, \mathbf{k}_{R,j}\rangle, \\ \Delta\mathbf{f}_{X,i} &= \mathcal{P}\left(\mathbf{v}_{R,j^*(i)}^c\right). \end{aligned} \tag{7}$$

The retrieved code value is projected and added to $\mathbf{f}_{X,i}^{\mathrm{attn}}$ as a residual, supplementing the continuous reference guidance. Flow guidance shapes joint attention and the features passed to subsequent blocks; codebook retrieval uses the shared queries and keys without the local attention mixture. This mechanism is applied in each DiT block as the joint context evolves.

To train the matching scores through hard reference retrieval, we use the original cover distribution $\boldsymbol{\alpha}^{\mathrm{soft}}$ from Eq. 4 to form the straight-through weights

$$\alpha_{ij}^{\mathrm{ste}} = \mathrm{sg}\big(\mathbb{I}[j = j^*(i)] - \alpha_{ij}^{\mathrm{soft}}\big) + \alpha_{ij}^{\mathrm{soft}}, \tag{8}$$

where $\mathbb{I}$ is the indicator function and sg stops gradients. During training, the retrieved value in Eq. 7 is evaluated as $\sum_j \alpha_{ij}^{\mathrm{ste}} \mathbf{v}_{R,j}^{c}$. The forward pass retains hard reference retrieval, while the softmax surrogate allows the restoration objective to update the matching queries and keys.

3.5 Training Strategy and Objectives

We first adapt the pretrained video DiT to cover-conditioned restoration using the in-context framework. We then introduce codebook injection and jointly fine-tune the backbone and the new projection modules, allowing reference matching and feature transfer to adapt together. The VAE, high-quality codebook, reference selector, and optical-flow estimator remain frozen during both stages. To account for imperfect native covers, training includes both clean and mildly degraded references.

We supervise the restored video with its high-quality target in latent and pixel spaces:

$$\mathcal{L} = \mathcal{L}_{\mathrm{lat}} + \mathcal{L}_{\mathrm{pix}} + \mathcal{L}_{\mathrm{per}} + \lambda_{\mathrm{temp}}\mathcal{L}_{\mathrm{temp}}. \tag{9}$$

The first two terms are mean squared reconstruction errors in latent and pixel space, and $\mathcal{L}_{\mathrm{per}}$ is the LPIPS perceptual loss. For temporal consistency, $\mathcal{L}_{\mathrm{temp}}$ penalizes bidirectional warping differences between adjacent restored frames using optical flow estimated from the target video, with occluded and out-of-bounds regions excluded. The weight λ_{temp} controls the temporal consistency term.

4 Experiments

4.1 Experimental Setup

Training data. We train on HQ-VSR (Chen et al., 2025) and generate low-quality inputs from high-quality videos using a two-stage degradation pipeline comprising blur, random resizing, noise, and compression, with a $4\times$ super-resolution scale. The middle frame serves as the reference, and training includes both clean and mildly degraded references to accommodate variations in native cover quality. The high-quality codebook and reference selector are pretrained on DF2K images.

Test datasets. We evaluate on the synthetic benchmarks **UDM10** (Tao et al., 2017), **SPMCS** (Yi et al., 2019), and **YouHQ40** (Zhou et al., 2024) at $4\times$ super-resolution. Each sequence uses a mildly degraded high-quality middle frame as its reference, with the high-quality video providing ground truth for reconstruction evaluation. For real-world evaluation, we construct **RealLive**, comprising 40 Live Photos captured with Xiaomi and iPhone devices. The dataset covers urban streets, architecture, natural landscapes, and indoor objects, with both daytime and nighttime captures. For each Live Photo, we restore the full recorded video using its original cover as the reference. Details of the synthetic benchmarks and the source Live Photo collection are provided in Appendix A.

Evaluation metrics. We evaluate restoration performance using image quality assessment (IQA) and video quality assessment (VQA) metrics. For IQA, we use PSNR and SSIM to evaluate reconstruction fidelity, and LPIPS (Zhang et al., 2018) and DISTS (Ding et al., 2020) to assess perceptual quality. Following LiveMoments (Xue et al., 2026), we also adopt four relative no-reference metrics: CLIP-IQA$_{\mathrm{re}}$, MUSIQ$_{\mathrm{re}}$, MANIQA$_{\mathrm{re}}$, and NIQE$_{\mathrm{re}}$. The relative score for a restored video is

$$q_{\mathrm{re}}(\widehat{Y}_{1:T}, R) = \frac{1}{T}\sum_{t=1}^{T} \frac{\Big|q(\widehat{Y}_t) - q(R)\Big|}{|q(R)|}, \tag{10}$$

where q is a no-reference quality metric and R the quality reference. Lower values indicate closer quality scores. We average relative errors over frames within each video, then equally over videos.

Table 1: Quantitative comparison with state-of-the-art methods. The best, second-best, and third-best results are highlighted in red, blue, and green, respectively. LiveVSR outperforms competing restoration methods across various datasets and metrics.

Dataset	Metric	RealBasicVSR	DOVE	SeedVR2	FlashVSR	SparkVSR*	SparkVSR†	ERVSR	LiveMoments	Live2K	LiveVSR (ours)
UDM10	PSNR↑	24.6086	26.5279	25.8315	23.6390	26.6219	23.4342	23.9667	24.7777	22.0786	29.1834
	SSIM↑	0.7008	0.7698	0.7472	0.7099	0.7757	0.6710	0.6694	0.7141	0.6401	0.8612
	LPIPS↓	0.3621	0.2710	0.2659	0.2798	0.2829	0.3550	0.5810	0.2576	0.5843	0.0960
	DISTS↓	0.1938	0.1519	0.1366	0.1449	0.1817	0.2036	0.3030	0.1238	0.2921	0.0547
	CLIP-IQA$_{re}$ ↓	0.3171	0.3235	0.3299	0.4117	0.1965	0.6865	0.5570	0.2240	0.3911	0.1924
	MUSIQ$_{re}$ ↓	0.0842	0.1003	0.2404	0.1485	0.1258	0.2158	0.6669	0.1281	0.6043	0.0245
	MANIQA$_{re}$ ↓	0.2073	0.2320	0.3277	0.4249	0.2039	0.6656	0.3988	0.1467	0.3008	0.0876
	NIQE$_{re}$ ↓	0.2123	0.2816	0.2661	0.1539	0.4246	0.1914	1.0036	0.1257	1.1253	0.0729
	DOVER↑	0.7651	0.7745	0.5571	0.8326	0.7102	0.8099	0.0945	0.6374	0.1208	0.7986
	E^*_{warp} ↓	2.1047	0.9823	1.4697	1.6615	0.7810	1.6279	3.8098	3.8118	4.1124	1.1053
SPMCS	PSNR↑	22.2829	23.0829	18.2038	21.3372	23.0436	20.1207	22.2151	23.7284	20.6603	27.9247
	SSIM↑	0.5656	0.6190	0.5377	0.5606	0.6238	0.4909	0.5653	0.7191	0.5353	0.8507
	LPIPS↓	0.3646	0.2887	0.4193	0.2806	0.3139	0.3388	0.6647	0.1458	0.6730	0.0729
	DISTS↓	0.2157	0.1710	0.2408	0.1584	0.1960	0.2022	0.3334	0.0935	0.3270	0.0618
	CLIP-IQA$_{re}$ ↓	0.1897	0.2293	0.3722	0.2300	0.1918	0.4162	0.3969	0.1497	0.3832	0.1392
	MUSIQ$_{re}$ ↓	0.0488	0.0508	0.2325	0.0558	0.0638	0.0651	0.6554	0.0568	0.6514	0.0249
	MANIQA$_{re}$ ↓	0.1415	0.1406	0.3125	0.1727	0.1693	0.2181	0.4633	0.1428	0.4247	0.0787
	NIQE$_{re}$ ↓	0.2364	0.5271	0.6481	0.1635	0.6585	0.3534	1.8723	0.1437	1.8866	0.1649
	DOVER↑	0.6824	0.7836	0.4757	0.7776	0.7265	0.7826	0.1212	0.7281	0.1282	0.7751
	E^*_{warp} ↓	1.4428	0.7771	6.1966	1.2838	0.6020	1.4456	3.0199	3.3931	3.8532	0.9610
YouHQ40	PSNR↑	22.4797	24.4441	18.7068	22.0383	24.3597	21.7559	23.7048	24.1908	21.8796	27.0240
	SSIM↑	0.5917	0.6808	0.6006	0.6057	0.6788	0.5786	0.6483	0.7184	0.6183	0.8039
	LPIPS↓	0.4082	0.2950	0.4176	0.2767	0.3273	0.3502	0.5566	0.1809	0.5775	0.1202
	DISTS↓	0.1935	0.1494	0.2102	0.1223	0.1872	0.1794	0.2634	0.0850	0.2578	0.0637
	CLIP-IQA$_{re}$ ↓	0.3181	0.3579	0.3647	0.5362	0.2983	0.6623	0.2974	0.1401	0.2736	0.1669
	MUSIQ$_{re}$ ↓	0.1146	0.1402	0.2705	0.1422	0.1837	0.1702	0.6187	0.0598	0.6096	0.0563
	MANIQA$_{re}$ ↓	0.2121	0.2597	0.3353	0.3394	0.2789	0.4275	0.3706	0.1083	0.3499	0.1324
	NIQE$_{re}$ ↓	0.2304	0.4236	0.4440	0.2124	0.6270	0.3336	1.5539	0.1002	1.6250	0.1309
	DOVER↑	0.8618	0.8513	0.7357	0.8941	0.8197	0.8585	0.3388	0.8259	0.3534	0.8708
	E^*_{warp} ↓	2.1294	1.1372	9.0816	2.4476	1.0108	1.8120	1.9092	4.3327	1.6483	1.5197
RealLive	CLIP-IQA$_{re}$ ↓	0.3241	0.2124	0.3012	0.2624	0.2139	0.3231	0.3040	0.1235	0.3123	0.1169
	MUSIQ$_{re}$ ↓	0.2257	0.1374	0.1763	0.2069	0.1235	0.1940	0.3612	0.0566	0.4148	0.0535
	MANIQA$_{re}$ ↓	0.2613	0.1562	0.1717	0.2538	0.1558	0.3293	0.2133	0.0664	0.1916	0.0969
	NIQE$_{re}$ ↓	0.3113	0.1901	0.1769	0.2962	0.1928	0.2176	0.3190	0.1314	0.4814	0.1208
	DOVER↑	0.7998	0.8020	0.7461	0.8233	0.7987	0.8187	0.6730	0.7754	0.6885	0.8200
	E^*_{warp} ↓	3.2863	2.8732	8.1145	6.1633	2.6936	3.6964	2.5769	5.2964	3.1397	3.9895

For VQA, DOVER (Wu et al., 2023) measures overall video quality, and flow warping error (Lai et al., 2018) measures temporal consistency. We report the scaled warping error $E^*_{\text{warp}} = 10^3 E_{\text{warp}}$.

On synthetic benchmarks, we report all metrics, using the high-quality video as ground truth and its clean middle frame as the reference for relative metrics. On RealLive, we report the four relative metrics, DOVER, and E^*_{warp}, using the native cover as the reference for relative metrics.

Implementation details. LiveVSR is initialized from the pretrained Wan2.1-T2V-1.3B (Wang et al., 2025a). We first train the reference-conditioned backbone with an initial learning rate of 2×10^{-5}, followed by joint fine-tuning with the codebook injection modules. Joint fine-tuning uses AdamW (Loshchilov & Hutter, 2019) with an effective batch size of 2. The VAE, codebook, and reference selector remain frozen throughout video training. At inference, we use overlapping spatiotemporal tiles and perform one restoration step per tile.

4.2 Comparison with State-of-the-Art Methods

Compared methods. We compare LiveVSR with the conventional VSR method RealBasicVSR (Chan et al., 2022b); diffusion-based VSR methods DOVE (Chen et al., 2025), SeedVR2 (Wang et al., 2026), FlashVSR (Zhuang et al., 2026), and SparkVSR (Yu et al., 2026); and reference-guided restoration methods ERVSR (Kim et al., 2023), LiveMoments (Xue et al., 2026), and Live2K (Lou et al., 2026). We evaluate SparkVSR in both reference-free (*) and PiSA-SR-guided (†) settings. We use the iPhone checkpoint of Live2K. LiveMoments is applied frame by frame, while Live2K uses a sliding temporal window; both use the same fixed reference throughout each sequence.

Quantitative results. We present quantitative comparisons in Table 1. LiveVSR achieves strong overall performance on both synthetic benchmarks and real-world Live Photos. For reconstruc-

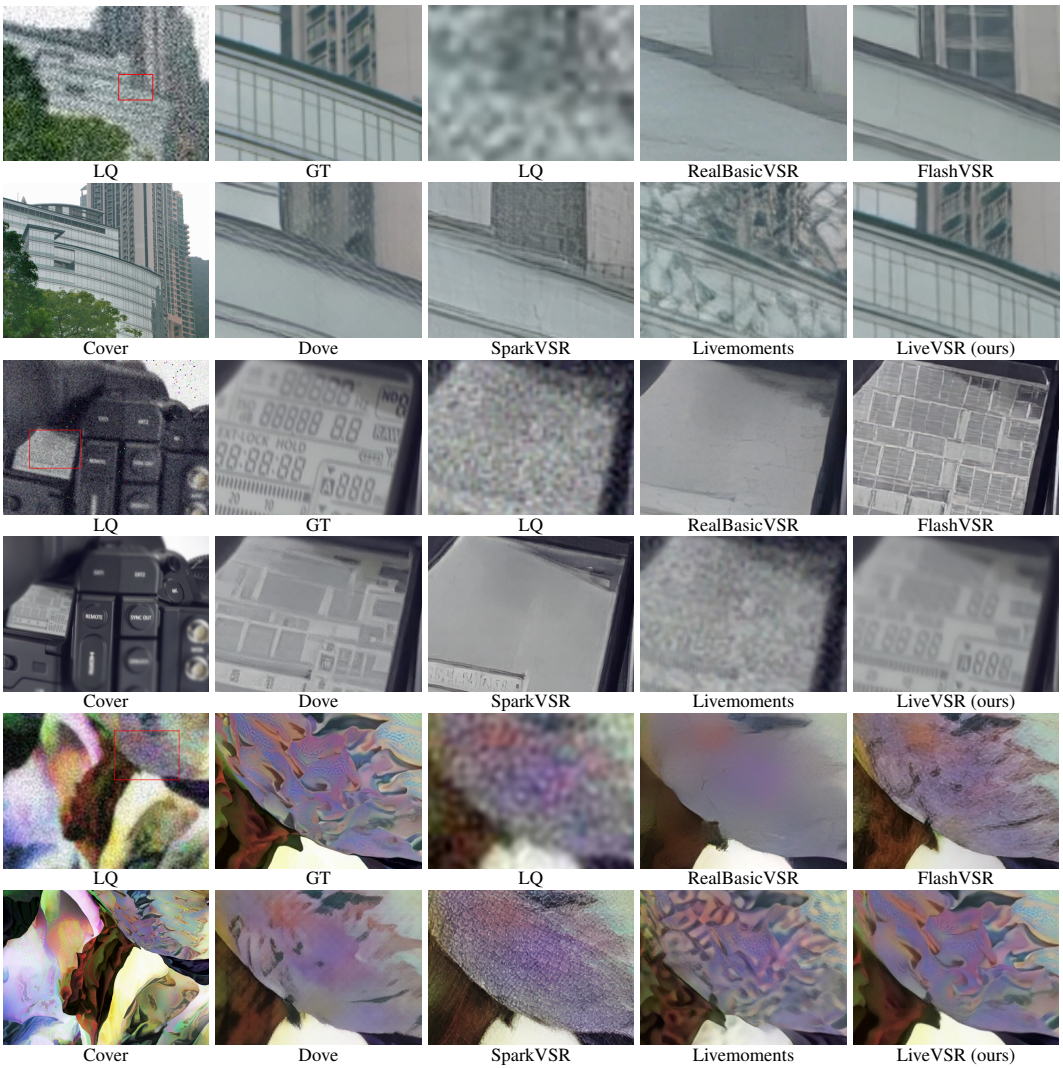


Figure 4: Qualitative comparison with state-of-the-art methods.

tion fidelity and perceptual quality, our method consistently outperforms competing methods across the synthetic datasets. On RealLive, it also achieves favorable cover-relative quality, with smaller quality-score discrepancies on most relative metrics. Furthermore, LiveVSR delivers competitive video quality and improves temporal consistency over frame-by-frame LiveMoments. These results demonstrate the effectiveness of our framework in restoring the full video sequence by using the native cover to guide the restoration of appearance details.

Qualitative comparisons. We present qualitative results on different scenes. Compared with existing methods, our approach restores sharper structures and more faithful fine-grained textures, while avoiding excessive smoothing and noticeable artifacts. The recovered details also better match the visual information in the cover image. Our outputs also agree more closely with the ground truth, particularly in regions containing high-frequency textures and small structures. We defer computational efficiency and temporal consistency analyses to Appendices C and D.

4.3 Ablation Studies

We investigate the effectiveness of the proposed components and reference-conditioning designs. All ablations use UDM10 and the 1.3B backbone. All models are evaluated on UDM10 using the 1.3B backbone, with configurations detailed in Appendix **??**. Results are presented in Tabs. 2 and 3, where the best and second-best scores are bold and underlined. Best and second-best results are bold and underlined; protocols and configuration differences are detailed in .

Table 2: Ablation study. (a) NoRef: no cover; +Ref/+Flow/+Codebook: cover conditioning/flow guidance/codebook injection. (b) Attention: cross-attention; RefNet: ReferenceNet; Ours: joint self-attention.

(a) Module ablation.

Metric	NoRef	+Ref	+Flow	+Codebook
PSNR↑	25.5923	28.7049	28.9483	**29.1833**
SSIM↑	0.7486	0.8523	0.8581	**0.8611**
DISTS↓	0.1648	0.0580	**0.0544**	0.0547
CLIP_{re} ↓	0.3402	0.1997	0.2141	**0.1924**
DOVER↑	0.7135	0.8007	**0.8033**	0.8014

(b) Reference utilization.

Metric	NoRef	Attention	RefNet	Ours
PSNR↑	25.5923	25.8977	26.2987	**28.7049**
SSIM↑	0.7486	0.7637	0.7683	**0.8523**
DISTS↓	0.1648	0.1939	0.1475	**0.0580**
CLIP_{re} ↓	0.3402	0.1997	**0.1667**	0.1997
DOVER↑	0.7135	0.5170	0.6441	**0.8007**

Table 3: Ablation study on UDM10. (a) Uni./Bi.: unidirectional/bidirectional interaction; +Flow: flow guidance. (b) Baseline: no codebook; Prompt: visual prompting; Full Attn.: full reference attention; Hard Retr.: Top-1 retrieval.

(a) In-context design.

Metric	Uni.	Bi.	Bi. +Flow
PSNR↑	28.4266	28.7049	**28.9483**
SSIM↑	0.8426	0.8523	**0.8581**
DISTS↓	0.0646	0.0580	**0.0544**
CLIP_{re} ↓	0.2165	**0.1997**	0.2141
DOVER↑	0.8005	0.8007	**0.8033**

(b) Codebook conditioning.

Metric	Baseline	Prompt	Full Attn.	Hard Retr.
PSNR↑	28.7049	28.8348	28.8276	**28.9623**
SSIM↑	0.8523	0.8564	0.8570	**0.8579**
DISTS↓	0.0580	0.0563	0.0561	**0.0556**
CLIP_{re} ↓	**0.1997**	0.2150	0.2153	0.2077
DOVER↑	0.8007	0.8051	**0.8053**	0.8045

Module ablation. Table 2a compares reference conditioning, flow guidance, and codebook injection. Cover conditioning provides the largest reconstruction gain, increasing PSNR by 3.11 dB over the baseline without a reference. Flow guidance further improves fidelity and perceptual quality by incorporating motion cues into reference matching. Adding codebook injection yields the highest PSNR and SSIM and the lowest CLIP_{re} error. These results support combining reference appearance, motion guidance, and discrete cover features for restoration.

Reference utilization. Table 2b compares cross-attention, ReferenceNet, and joint self-attention. Joint self-attention updates cover and video representations together within a shared context. It achieves the highest PSNR, SSIM, and DOVER and the lowest DISTS, although ReferenceNet gives a lower CLIP_{re} error. The substantial fidelity and perceptual gains support jointly modeling cover and video features to use reference details more effectively.

In-context design. Without codebook injection, Tab. 3a compares unidirectional and bidirectional designs. Unidirectional interaction restricts cover tokens to cover information. Bidirectional interaction instead updates cover features with video context, improving PSNR from 28.43 to 28.70 dB. Flow guidance further raises PSNR to 28.95 dB and reduces DISTS to 0.0544. These gains suggest that motion cues provide useful spatial guidance beyond content-based reference matching.

Codebook conditioning. We compare visual prompting, full reference attention, and hard reference retrieval in Tab. 3b. All three improve reconstruction over the baseline without codebook injection. Hard retrieval selects one cover location for each video token, limiting the mixing of discrete features from different locations.

5 Conclusion

In this paper, we formulate the task of cover-guided Live Photo super-resolution and propose LiveVSR, a diffusion-based framework that restores the complete video sequence using the native cover. We introduce an in-context framework with bidirectional cover-video interaction and flow-guided local retrieval to establish reliable reference correspondences. Furthermore, correspondence-guided codebook injection complements continuous reference features with a discrete visual prior from high-quality images. Experiments on synthetic benchmarks and real-world Live Photos demonstrate that LiveVSR achieves superior reconstruction fidelity and perceptual quality, together with competitive video quality and temporal consistency.

AI USE STATEMENT

In this work, we used generative AI tools to assist with translation, manuscript drafting, language polishing, and literature searches. These tools also supported code implementation, the presentation of mathematical notation, the organization and interpretation of experimental results, and the preparation of tables and LaTeX source. All AI-assisted code and manuscript text were reviewed by the authors for accuracy and consistency with the method descriptions and experimental records. We take full responsibility for the final content of this work, including its text, technical claims, experimental findings, and artifacts produced with the assistance of generative AI.

ETHICS STATEMENT

Our work studies cover-guided Live Photo restoration for mobile photography. RealLive was collected with participants' consent for data collection and public release. Identifiable individuals will be removed or anonymized before release to protect privacy. Generative restoration may introduce inaccurate details, and the outputs should not be treated as verified records of the original scene.

REPRODUCIBILITY STATEMENT

The model and training objectives are described in Sec. 3. Training and inference settings are provided in Sec. 4.1, with dataset details in Appendix A and evaluation protocols in Appendix B. Computational efficiency is reported in Appendix C. We plan to release our code and the RealLive dataset to support reproducibility and further research.

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A Dataset Details

A.1 Training Data Preparation

Each HQ-VSR training sample contains 17 consecutive frames at a target resolution of 256×448. We synthesize the LQ video using the two-stage degradation pipeline described in Sec. 4.1, with a $4\times$ super-resolution scale, and select the high-quality middle frame as the reference source. We retain this reference without degradation with probability 0.1 and apply mild degradation with strength 0.10 with probability 0.9. This mixture exposes the model to both clean and imperfect cover observations.

A.2 RealLive

Dataset composition. RealLive comprises 40 Live Photos: 21 captured with a Xiaomi REDMI K80 and 19 with iPhone devices. Table 4 summarizes the device and cover-resolution groups. The videos contain 7–90 frames each, totaling 2912 frames across the dataset. The cover resolutions in the table refer to the original photographs before evaluation crops are extracted.

Table 4: Statistics of RealLive grouped by capture device and original cover resolution.

Device	Cover resolution	Live Photos
Xiaomi REDMI K80	12 MP	21
iPhone	12 MP	5
iPhone	24 MP	14
Total	—	40

Spatial preprocessing. For evaluation, we use spatial crops of RealLive. Each cover is center-cropped to 2560×1920 pixels, or 1920×2560 for portrait captures. The associated video is center-cropped using the corresponding spatial proportions, with dimensions rounded to an even-sized rectangle that retains its original aspect ratio. No resizing is applied when constructing these crops, and no synthetic degradation is added to the real inputs.

Sequence and reference protocol. We keep the cover fixed throughout restoration and restore every video frame to the cropped cover's resolution. The original frame count, temporal order, and presentation timestamps are preserved. The cover provides appearance guidance and a quality reference; it is not used as a frame-aligned reconstruction target for the video.

A.3 Synthetic Benchmarks

On UDM10, SPMCS, and YouHQ40, we use the low-quality videos as inputs and the corresponding high-quality sequences as reconstruction targets for $4\times$ super-resolution. For reference-guided methods, we apply mild degradation with strength 0.10 to the high-quality middle frame and use the resulting image as a fixed reference for the entire sequence. For the cover-relative quality metrics, the quality reference is the clean high-quality middle frame, as specified in Sec. 4.1.

B Evaluation Details

Relative metrics. Following LiveMoments (Xue et al., 2026), we compute the relative score for a restored video as

$$q_{\mathrm{re}}(\widehat{Y}_{1:T}, R) = \frac{1}{T}\sum_{t=1}^{T} \frac{\left|q(\widehat{Y}_t) - q(R)\right|}{|q(R)|}, \tag{11}$$

where q is CLIP-IQA, MUSIQ, MANIQA, or NIQE, and R is the quality reference. For RealLive, R is the native cover; for the synthetic benchmarks, it is the clean high-quality middle frame. We

average the frame-level discrepancies within each video, then average equally across videos to obtain the dataset score.

C Computational Efficiency

Table 5 summarizes the recorded computational cost on UDM10, comprising 10 videos with 32 frames each at an output resolution of 720×1272 for $4\times$ super-resolution. We report the recorded total processing time divided by 10 as seconds per video, together with the peak GPU memory and complete inference parameter count. Parameter counts include frozen inference modules, count shared weights once, and exclude buffers, cached input embeddings, and evaluation networks. LiveMoments and our method use the mildly degraded middle-frame reference; SparkVSR uses a PiSA-SR-generated reference, and its parameter count includes both models.

Table 5: Computational efficiency on UDM10. Parameters cover the complete inference pipeline ($\mathrm{M} = 10^6$). Runtime and memory retain the measurement conventions of the original runs, as detailed below.

Method	Parameters (M)	Time (s/video)	Peak memory (GiB)
DOVE	10548.57	28.79	26.56
LiveMoments	4268.45	132.80	11.51
FlashVSR	1780.14	15.10	22.20
SparkVSR	11839.44	42.75	37.35[b]
Ours	1689.78	20.59[a]	19.98[a]

Measurement scope. DOVE and FlashVSR timings include model loading, inference, and video encoding. LiveMoments sums per-frame inference and PNG processing time, excluding model loading, input extraction, video encoding, and metric evaluation. The retained runtime for our earlier pipeline includes inference and video saving, but excludes model loading, input preparation, warm-up, and metric evaluation. Timing boundaries and memory statistics differ across the original runs, so these figures do not constitute a comparison under a uniform profiling protocol.

D More Visual Results

In Figs. 5, **??**, and 6, we present additional visual comparisons across different scenes and datasets. The results include enlarged regions containing fine-grained textures and small structures for clearer comparison with competing methods. These examples further demonstrate that LiveVSR recovers sharper structures and more faithful details while maintaining consistent visual quality across diverse scenarios.

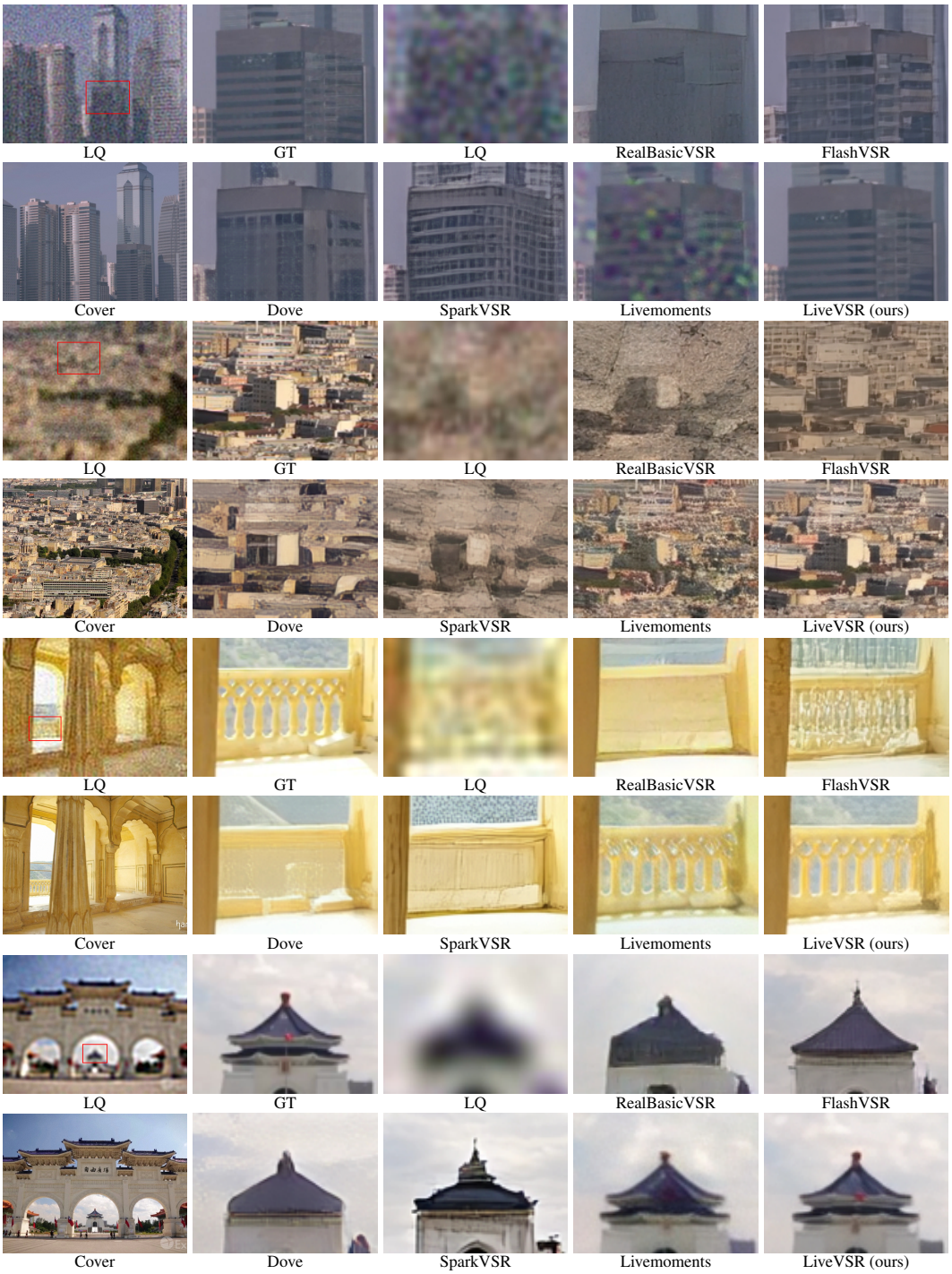


Figure 5: Qualitative comparison with state-of-the-art methods.

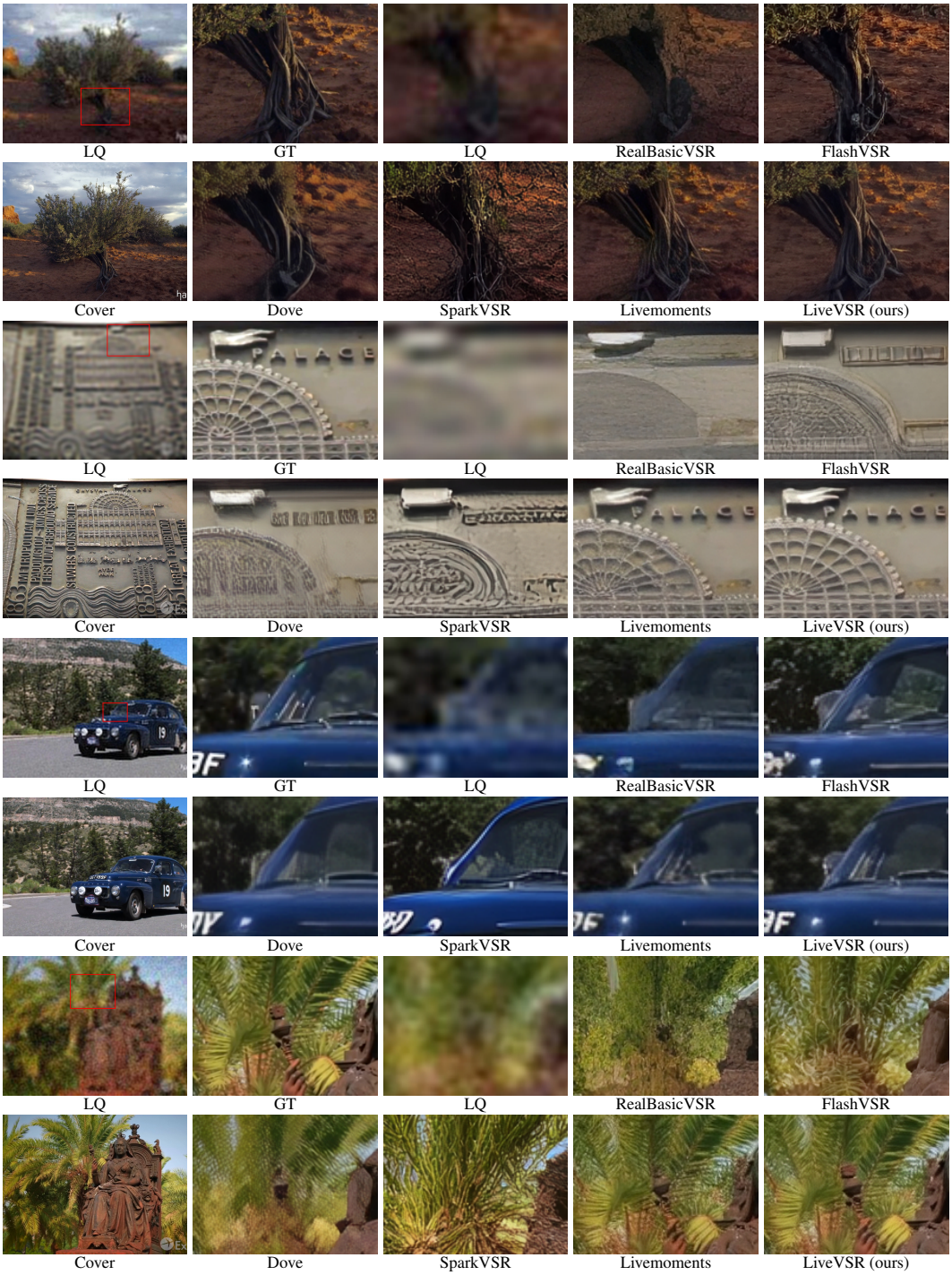


Figure 6: Qualitative comparison with state-of-the-art methods.